CENTROID GEOMETRY FOR EXPONENTIAL SUMS OVER ZETA ORDINATES AND LAMBERT-APPROXIMATED ORDINATES

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Remark 0.1. This article is an exploratory numerical notebook. It records computations from the CentroidGeometry project generated on 08-Jun-2026. The statements below are numerical observations unless explicitly identified otherwise. No implication for the Riemann hypothesis is claimed. No proof of a new theorem is claimed.

1. Introduction

The standard background object is the sequence of nontrivial zeros of the Riemann zeta function [1, 2]. The present notebook studies only the ordinates used by the project data files. Let these ordinates be denoted by

$$\gamma_1, \gamma_2, \ldots, \gamma_N. \tag{1.1}$$

The project also uses a Lambert-approximated ordinate sequence

$$\lambda_1, \lambda_2, \ldots, \lambda_N, \tag{1.2}$$

whose numerical construction is part of the generated project workflow. The Lambert function is a standard special function [3].

Definition 1.1. For the zeta-ordinate sequence, define

$$S_n = \sum_{k \le n} \exp(i\gamma_k), \tag{1.3}$$

and

$$C_n = \frac{1}{n} \sum_{m \le n} S_m. \tag{1.4}$$

For the Lambert-approximated sequence, define

$$S_n^{(L)} = \sum_{k \le n} \exp(i\lambda_k), \tag{1.5}$$

and

$$C_n^{(L)} = \frac{1}{n} \sum_{m \le n} S_m^{(L)}. \tag{1.6}$$

Remark 1.2. The generated reports detect

$$N = 2001052 \tag{1.7}$$

for both centroid sequences in the comparisons. All comparisons below use this common length.

2. Project Sources and Numerical Status

The original project material is contained in the generated sourcebook `CG_SOURCEBOOK_20260608_134200.txt`. The numerical values quoted below are taken from the corresponding generated reports in `results/tables/`. The figures are the generated PNG files in `results/figures/`.

Remark 2.1. The use of (1.3)–(1.6) is definitional. The numerical ranges, means, standard deviations, and visual descriptions are original project observations. The article does not include a heuristic derivation of the observed geometry, and it does not state any conjecture.

3. Full-Sequence Evolution of the Zeta Centroid

The full-sequence reports CG20 and CG21 record the coordinate evolution of C_n over all $N = 2001052$ values. For the real coordinate, the generated report gives

$$\operatorname{mean} \operatorname{Re}(C_n) = 0.776915630527036, \tag{3.1}$$

$$\operatorname{std} \operatorname{Re}(C_n) = 1.430501040663642 \cdot 10^{-3}, \tag{3.2}$$

with minimum -0.280575369521360, maximum 1.115627993757473, and final value 0.776907119554484. For the imaginary coordinate, the report gives

$$\operatorname{mean} \operatorname{Im}(C_n) = 0.767874106258068, \tag{3.3}$$

$$\operatorname{std} \operatorname{Im}(C_n) = 1.584640594307709 \cdot 10^{-3}, \tag{3.4}$$

with minimum 0.647012631360417, maximum 1.526375113112613, and final value 0.767844590760645.

4. First One Thousand Values

The first 1000 values show the initial transient visible before the later compressed scale dominates the full-sequence figures. For $\operatorname{Re}(C_n)$, CG22 reports

$$\operatorname{mean}_{1 \le n \le 1000} \operatorname{Re}(C_n) = 0.775562350234487, \tag{4.1}$$

with standard deviation $6.361888696782327 \cdot 10^{-2}$, minimum -0.280575369521360, maximum 1.115627993757473, first value 0.002441796727579, and last plotted value 0.772074257902768. For $\operatorname{Im}(C_n)$, CG23 reports

$$\operatorname{mean}_{1 \le n \le 1000} \operatorname{Im}(C_n) = 0.783259112250021, \tag{4.2}$$

with standard deviation $6.885194697768009 \cdot 10^{-2}$, minimum 0.647012631360417, maximum 1.526375113112613, first value 0.999997018809927, and last plotted value 0.762675411041596.

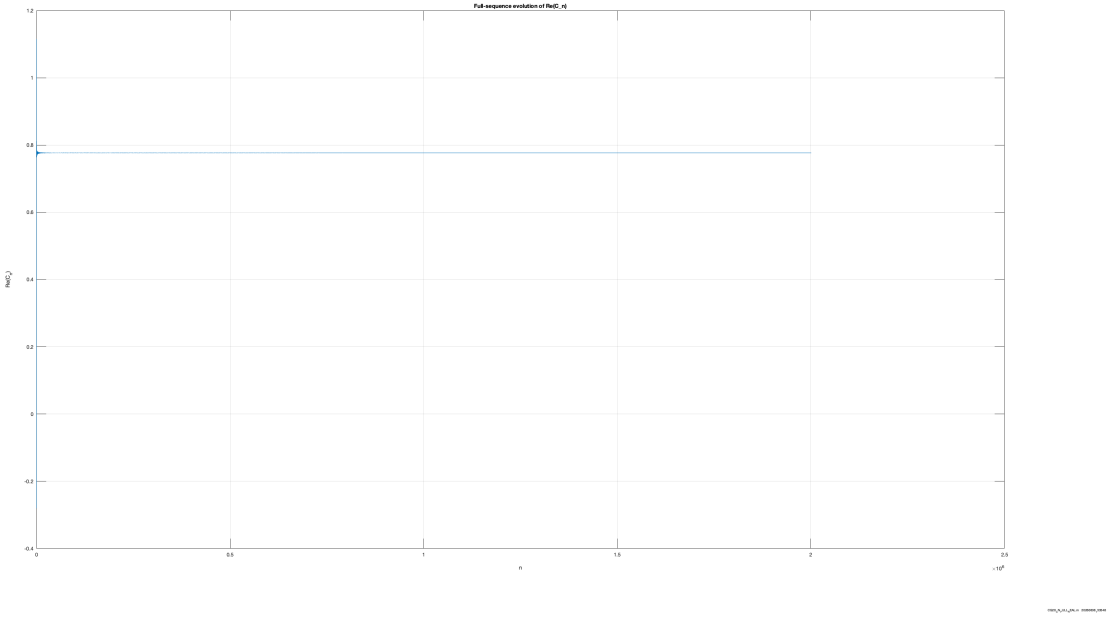


FIGURE 1. Full-sequence evolution of $\mathrm{Re}(C_n)$, generated by CG20.

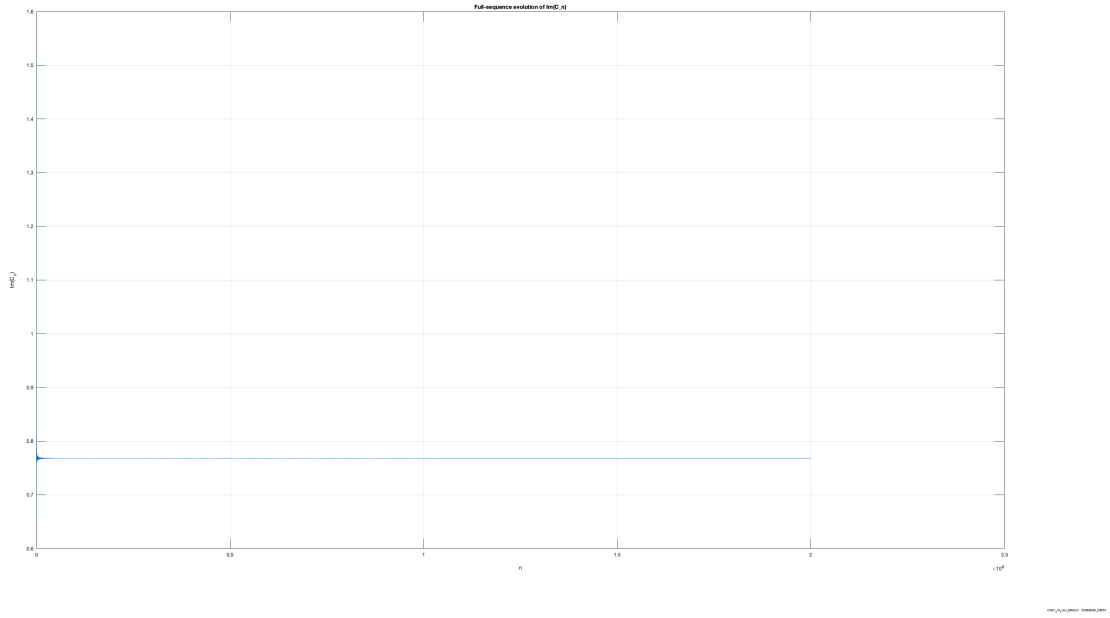


FIGURE 2. Full-sequence evolution of $\mathrm{Im}(C_n)$, generated by CG21.

5. BEHAVIOUR AFTER $n = 10^4$

The reports CG24 and CG25 isolate the range

$$10000 \leq n \leq 2001052. \tag{5.1}$$

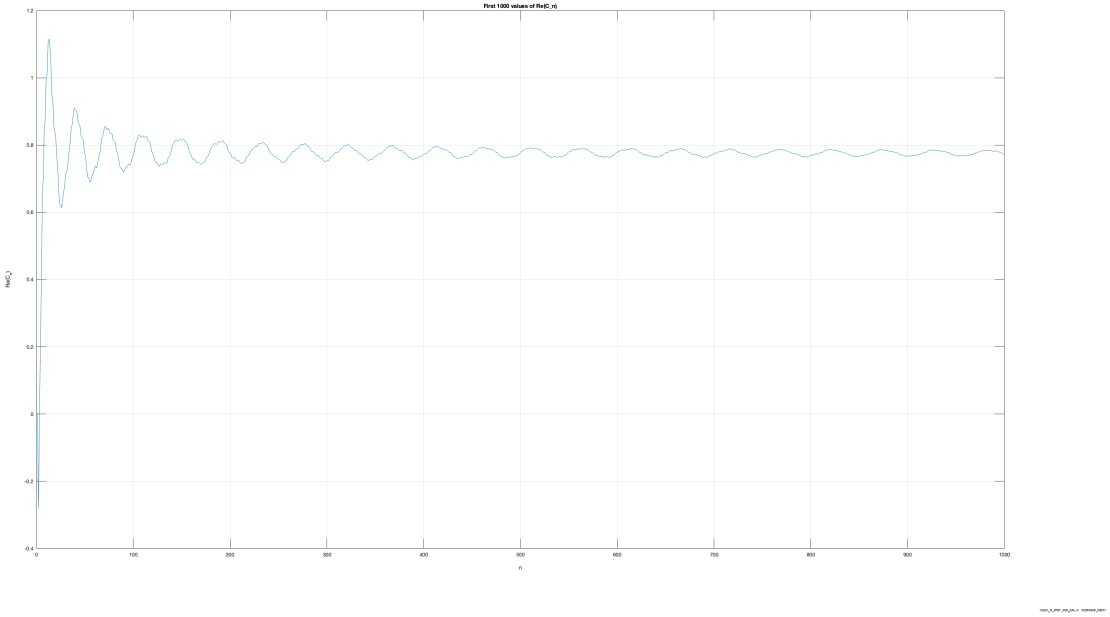


FIGURE 3. First 1000 values of $\mathrm{Re}(C_n)$, generated by CG22.

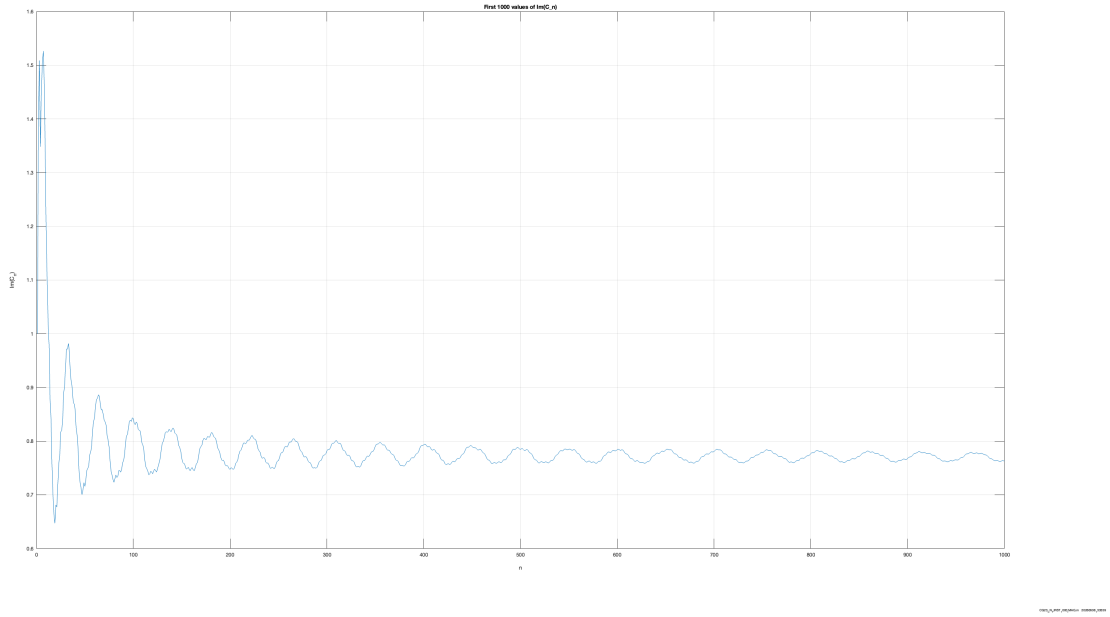


FIGURE 4. First 1000 values of $\mathrm{Im}(C_n)$, generated by CG23.

There are 1991053 plotted points in this range. For the real coordinate, CG24 reports mean 0.776917475084998, standard deviation $6.794344393288941 \cdot 10^{-5}$, minimum 0.775485220001544, and maximum 0.778119395416095. For the imaginary coordinate, CG25 reports mean 0.767864442040498, standard deviation $7.350593637830189 \cdot 10^{-5}$, minimum 0.766837649133883,

and maximum 0.769478180509284. Numerically, this post-10^4 window is much narrower than the first 1000 window reported in Section 4.

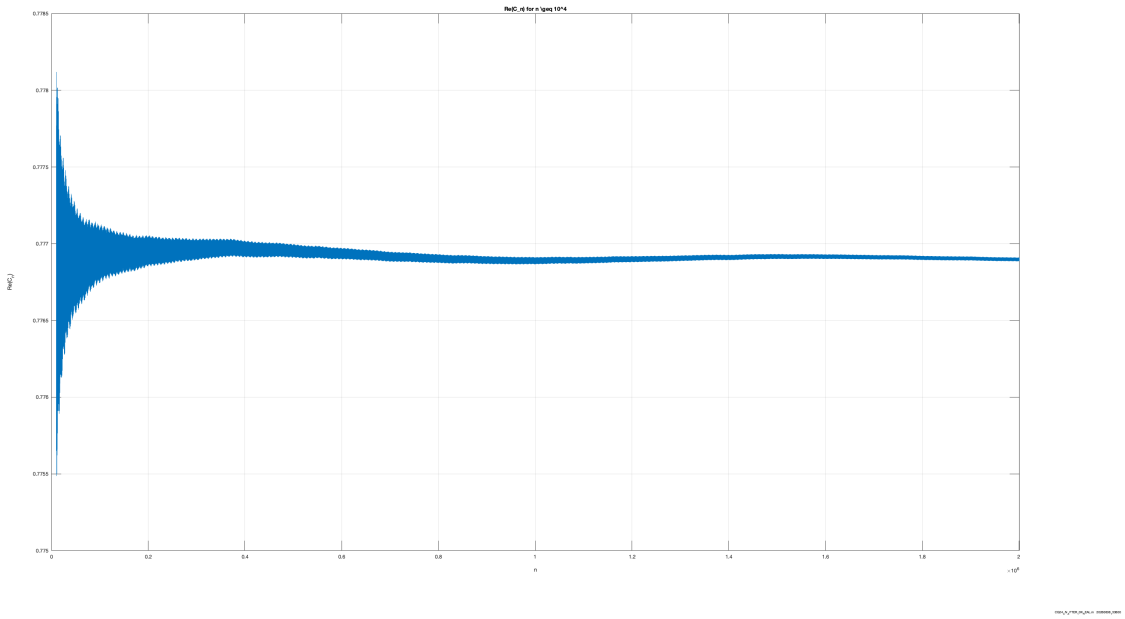


FIGURE 5. Evolution of $\mathrm{Re}(C_n)$ for $n \geq 10^4$, generated by CG24.

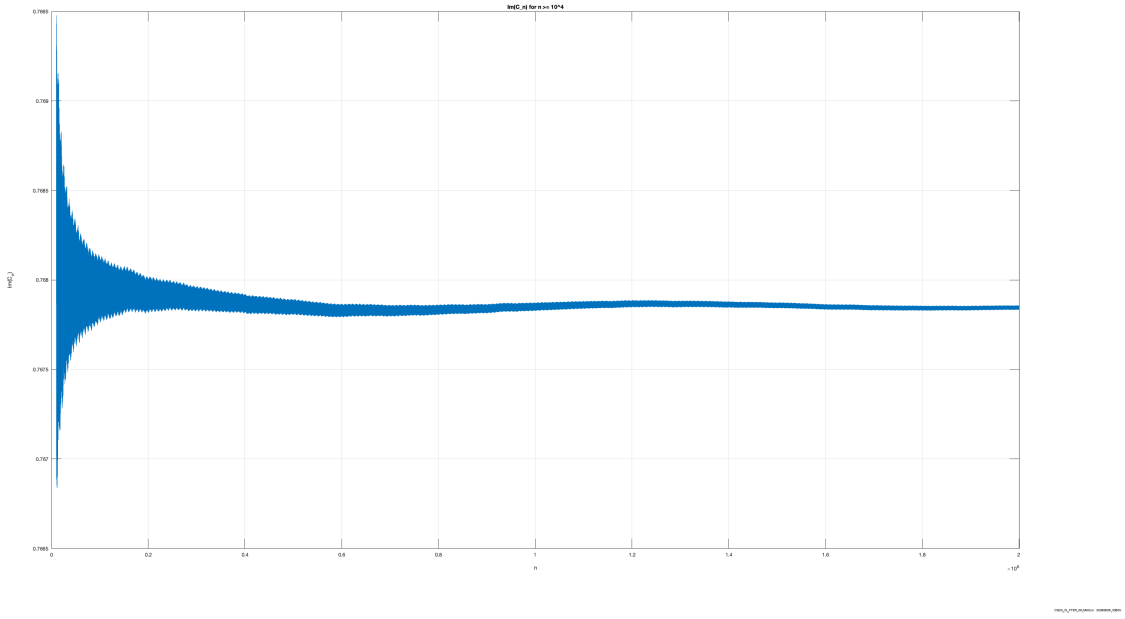


FIGURE 6. Evolution of $\mathrm{Im}(C_n)$ for $n \geq 10^4$, generated by CG25.

6. Complex Cloud and 3D Evolution of C_n

The complex cloud plot records the points

$$(\mathrm{Re}(C_n), \mathrm{Im}(C_n)). \tag{6.1}$$

CG26 reports real-coordinate mean 0.776915630527036 and standard deviation 0.001430501040664, and imaginary-coordinate mean 0.767874106258068 and standard deviation 0.001584640594308. The 3D plot records the same complex coordinates against n. CG27 reports the coordinate ranges

$$-0.280575369521360 \le \mathrm{Re}(C_n) \le 1.115627993757473, \tag{6.2}$$

and

$$0.647012631360417 \le \mathrm{Im}(C_n) \le 1.526375113112613. \tag{6.3}$$

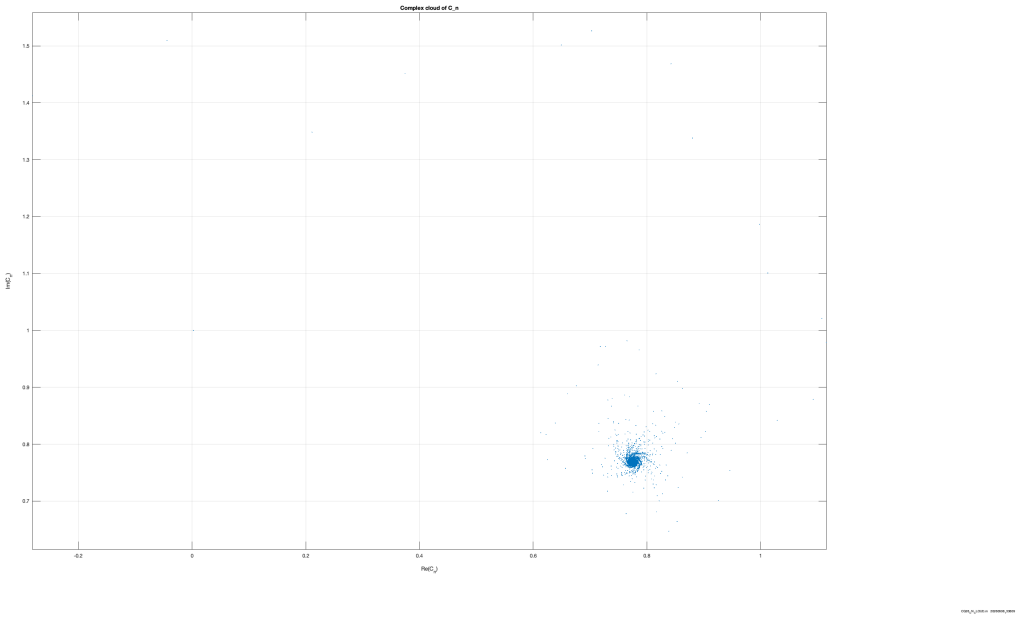


Figure 7. Complex cloud of C_n, generated by CG26.

7. Lambert Centroid Cloud and 3D Evolution

The Lambert centroid sequence $C_n^{(L)}$ is examined using the same cloud and 3D views. CG30 reports real-coordinate mean 0.041563217656466 and standard deviation 0.000592355552691, and imaginary-coordinate mean 1.757537205811400 and standard deviation 0.000670139062160. CG31 reports

$$-0.491409250871176 \le \mathrm{Re}(C_n^{(L)}) \le 0.041681041002623, \tag{7.1}$$

and

$$0.927106075076750 \le \mathrm{Im}(C_n^{(L)}) \le 1.757937547844233. \tag{7.2}$$

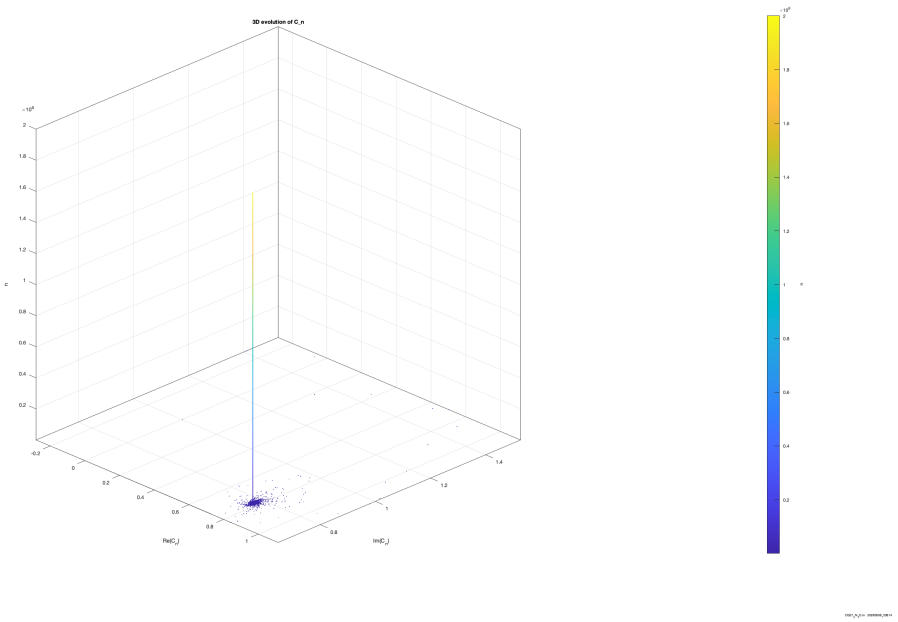


FIGURE 8. 3D evolution of C_n with n as vertical coordinate, generated by CG27.

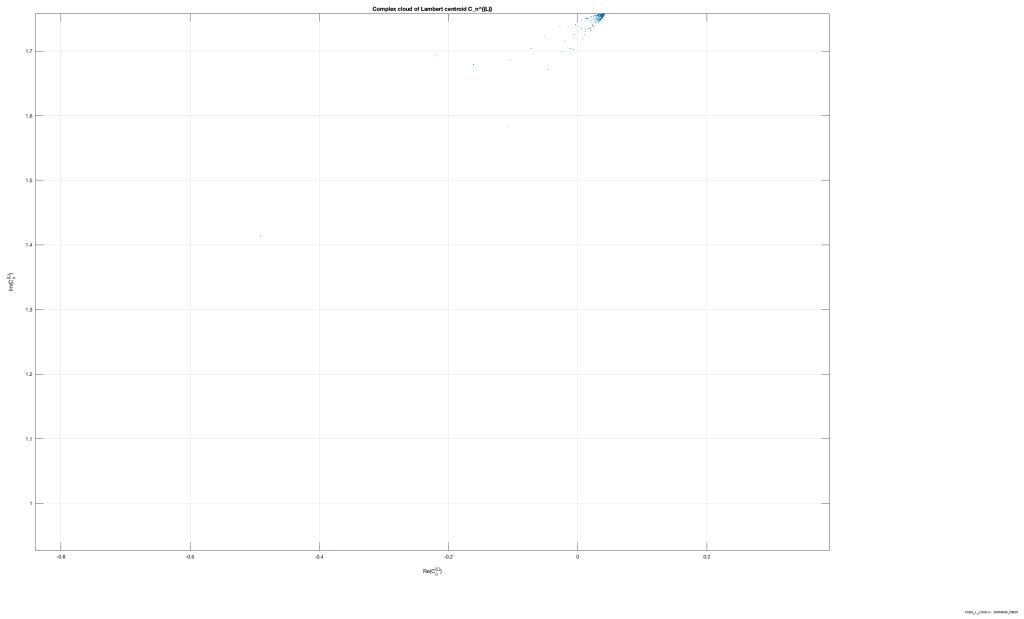


FIGURE 9. Complex cloud of $C_n^{(L)}$, generated by CG30.

8. CENTERED COMPARISON

The centered comparison uses the final tenth of the available index range:

$$I_{\text{tail}} = \{\lfloor 0.9N \rfloor, \ldots, N\}. \tag{8.1}$$

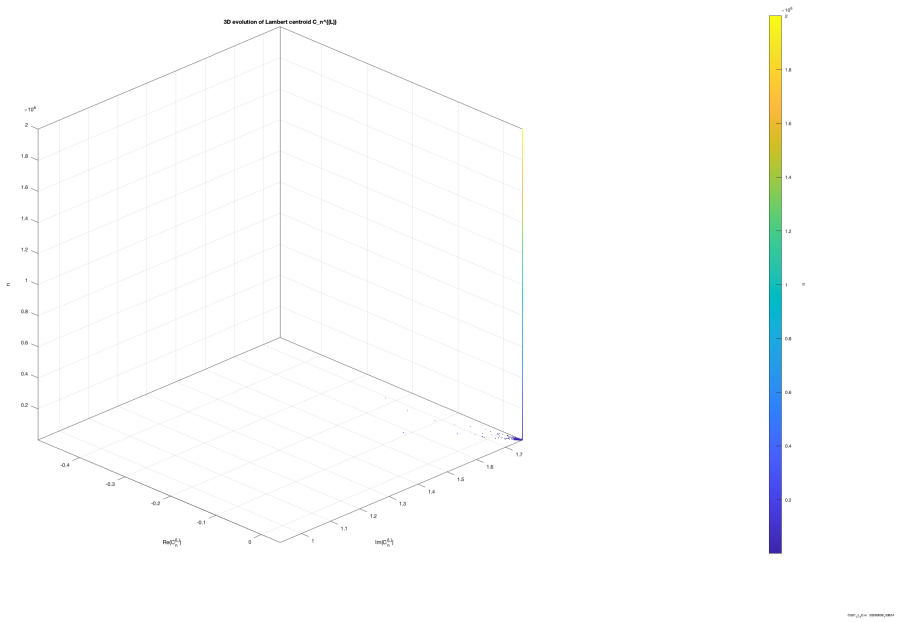


FIGURE 10. 3D evolution of $C_n^{(L)}$, generated by CG31.

The centers used by the generated script are

$$C_0 = \frac{1}{|I_{\mathrm{tail}}|} \sum_{n \in I_{\mathrm{tail}}} C_n \tag{8.2}$$

and

$$C_0^{(L)} = \frac{1}{|I_{\mathrm{tail}}|} \sum_{n \in I_{\mathrm{tail}}} C_n^{(L)}. \tag{8.3}$$

The centered sequences are

$$\widetilde{C}_n = C_n - C_0 \tag{8.4}$$

and

$$\widetilde{C}_n^{(L)} = C_n^{(L)} - C_0^{(L)}. \tag{8.5}$$

CG40 reports $N = 2001052$ for this comparison.

9. Normalized Comparison

The normalized comparison rescales the centered sequences by their mean tail radii:

$$r_C = \frac{1}{|I_{\mathrm{tail}}|} \sum_{n \in I_{\mathrm{tail}}} |\widetilde{C}_n| \tag{9.1}$$

and

$$r_L = \frac{1}{|I_{\mathrm{tail}}|} \sum_{n \in I_{\mathrm{tail}}} |\widetilde{C}_n^{(L)}|. \tag{9.2}$$

The normalized sequences are

$$\widehat{C}_n = \frac{\widetilde{C}_n}{r_C} \tag{9.3}$$

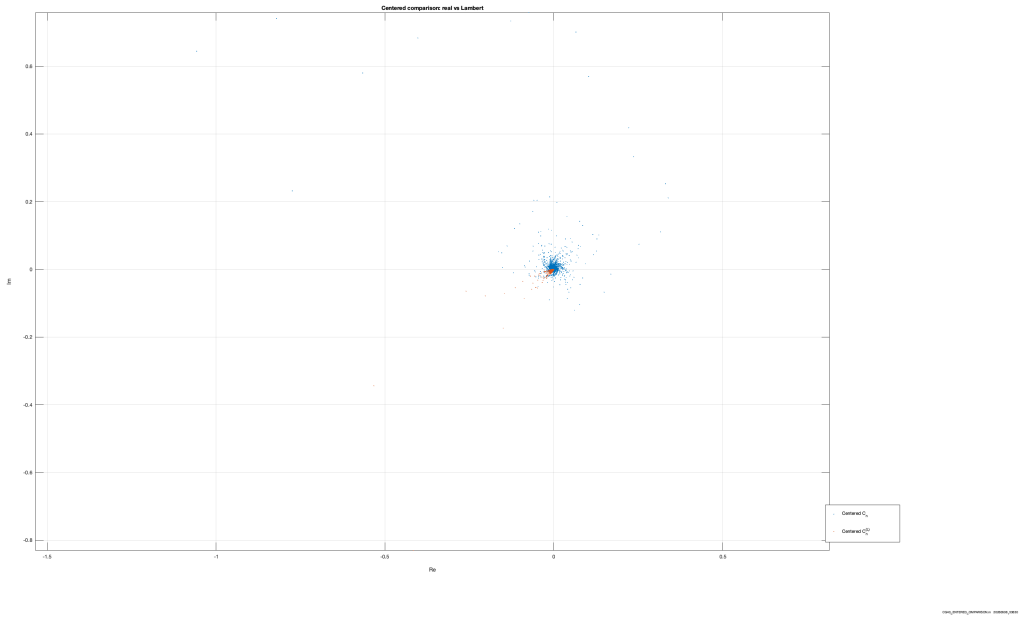


FIGURE 11. Centered comparison between C_n and $C_n^{(L)}$, generated by CG40.

and

$$\widehat{C}_n^{(L)} = \frac{\widetilde{C}_n^{(L)}}{r_L}. \tag{9.4}$$

CG41 reports

$$r_C = 0.000009963352826 \tag{9.5}$$

and

$$r_L = 0.000001982288500. \tag{9.6}$$

10. Difference Sequence D_n

The generated difference sequence is formed after centering and normalizing:

$$D_n = \widehat{C}_n - \widehat{C}_n^{(L)}. \tag{10.1}$$

CG50 reports

$$\text{mean}\,|D_n| = 9.931235651811170 \tag{10.2}$$

and

$$\text{std}\,|D_n| = 485.727132746548079. \tag{10.3}$$

CG51 reports the 3D coordinate ranges

$$-3854.387908394064652 \le \text{Re}(D_n) \le 162733.198123055801261 \tag{10.4}$$

and

$$-3312.321233199529161 \le \text{Im}(D_n) \le 442227.983543796406593. \tag{10.5}$$

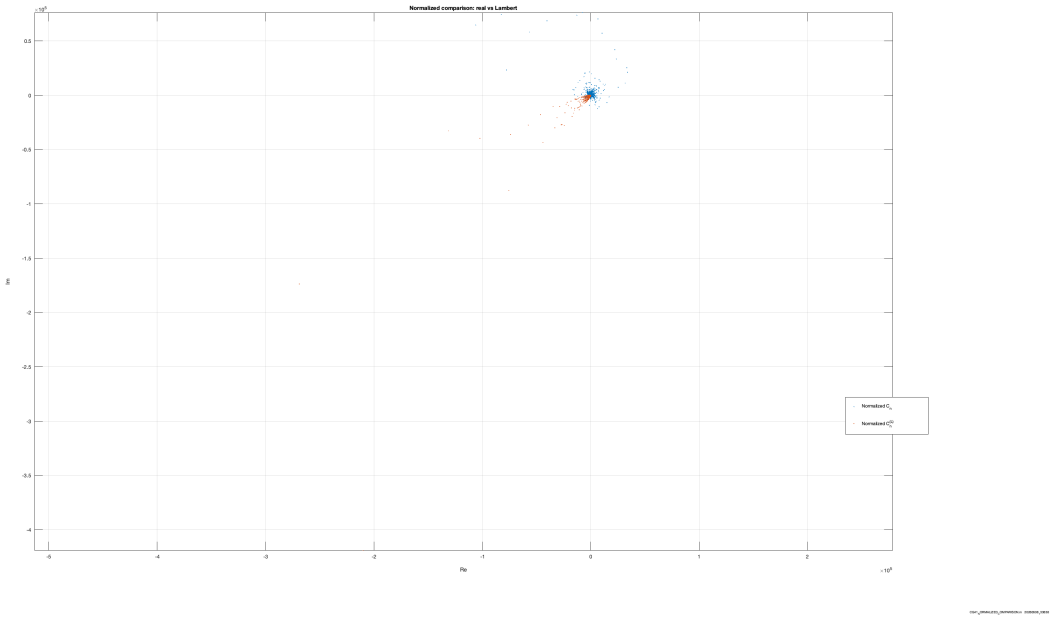


FIGURE 12. Normalized comparison between C_n and $C_n^{(L)}$, generated by CG41.

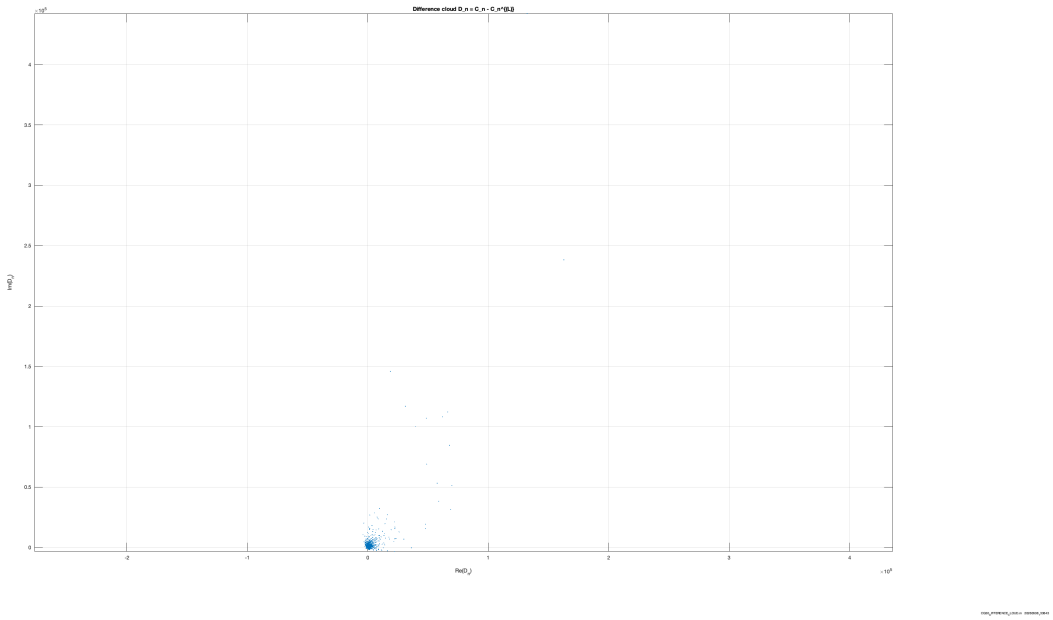


FIGURE 13. Complex cloud of D_n, generated by CG50.

11. RATIO SEQUENCE R_n

The generated ratio is also formed after centering and normalizing. For indices satisfying the denominator threshold

$$|\widehat{C}_n^{(L)}| > 10^{-12}, \tag{11.1}$$

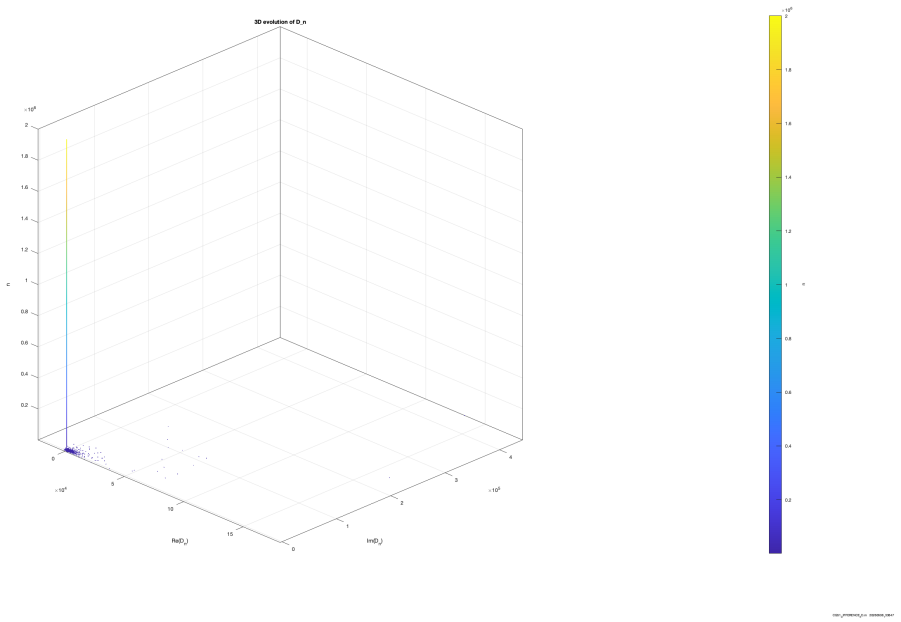


FIGURE 14. 3D evolution of D_n, generated by CG51.

the ratio is

$$R_n = \frac{\widehat{C}_n}{\widehat{C}_n^{(L)}}. \tag{11.2}$$

CG60 reports 2001052 valid ratio points, with

$$\operatorname{mean} |R_n| = 1.507164923130925 \tag{11.3}$$

and

$$\operatorname{std} |R_n| = 0.824760433136343. \tag{11.4}$$

The full 3D ratio view in CG62 reports the same number of valid points and radius range

$$0.002226938425381 \leq |R_n| \leq 333.143351039707454. \tag{11.5}$$

CG61 restricts the 3D view to the last tenth of the index range and reports 200107 points, radius mean 1.000216091805182, and radius standard deviation 0.276011678349894.

12. Filtered Ratio Cylinder

The filtered ratio cylinder retains finite ratio values satisfying

$$|R_n| < 10. \tag{12.1}$$

CG63 reports $N = 2001052$ total indices and 2000935 filtered points. For the filtered set,

$$\operatorname{mean} |R_n| = 1.505986471281209, \tag{12.2}$$

$$\operatorname{std} |R_n| = 0.770419340022107, \tag{12.3}$$

with minimum 0.002226938425381 and maximum 9.972059492555911.

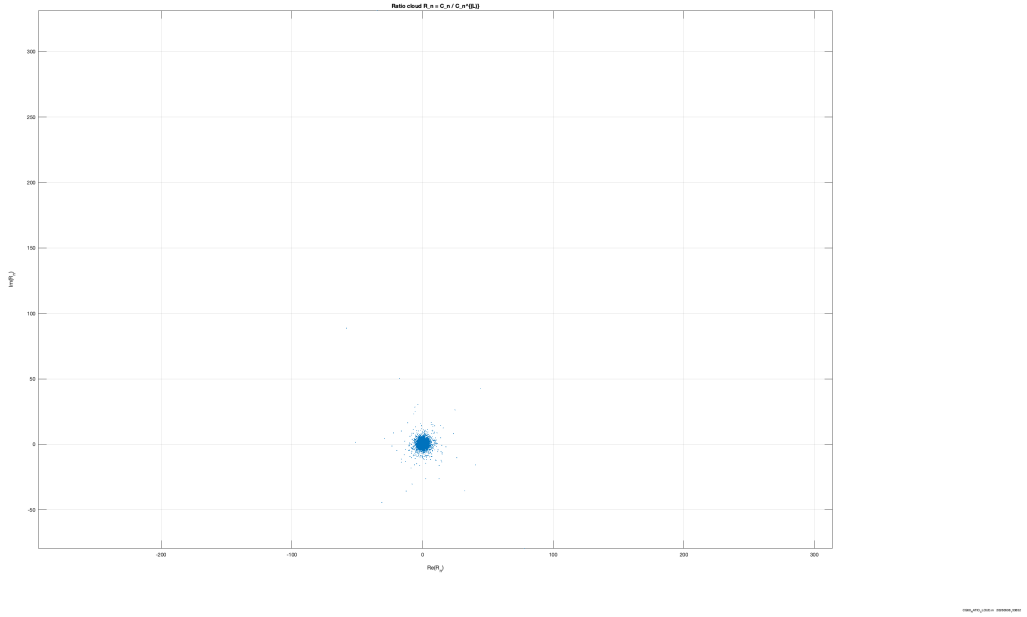


FIGURE 15. Complex cloud of R_n, generated by CG60.

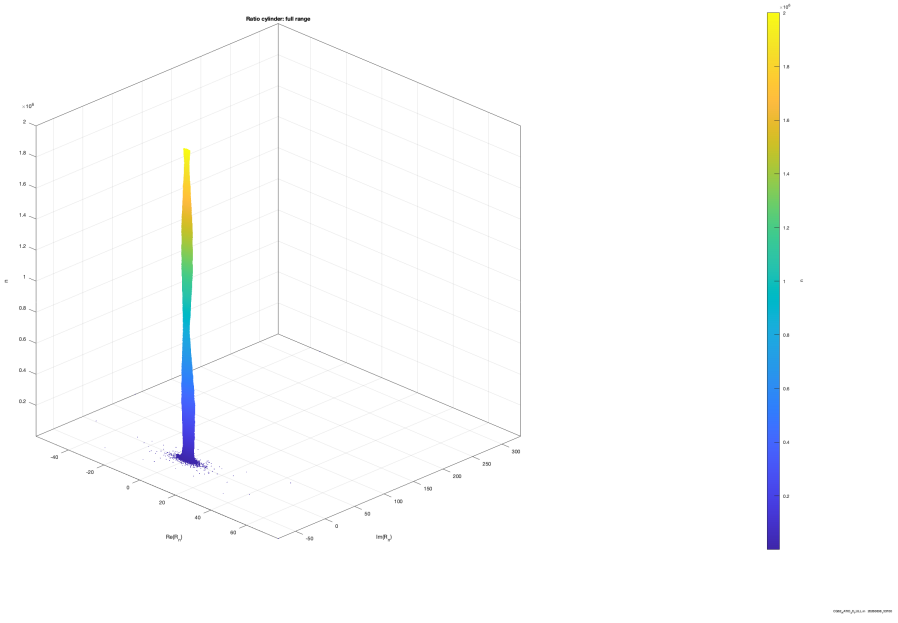


FIGURE 16. Full 3D evolution of R_n, generated by CG62.

13. Radius, Thickness, and Envelope of the Ratio

The radius sequence is

$$\rho_n = |R_n|. \tag{13.1}$$

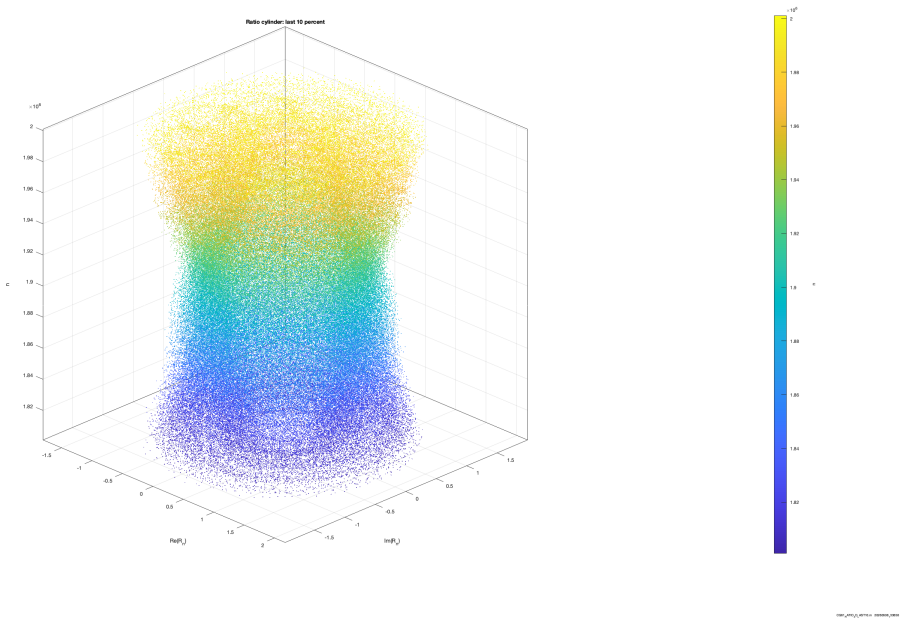


FIGURE 17. 3D evolution of R_n on the last 10% of indices, generated by CG61.

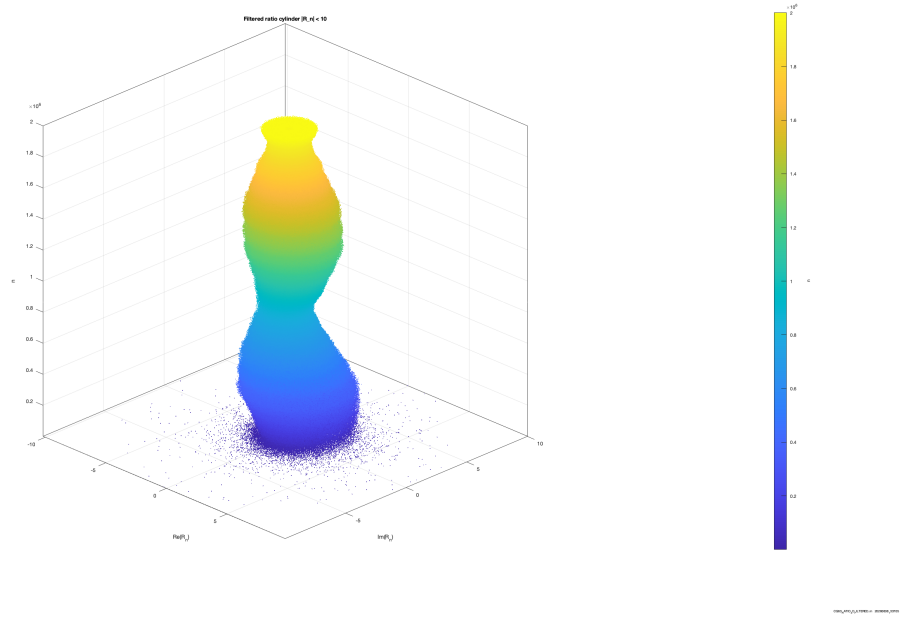


FIGURE 18. Filtered ratio cylinder satisfying $|R_n| < 10$, generated by CG63.

CG70 plots this radius evolution. For the filtered cylinder, CG71 computes a moving standard deviation over the window

$$W = 10000. \tag{13.2}$$

The reported local thickness statistics are mean 0.658810631388686, standard deviation 0.212717848419338, minimum 0.173106476121596, and maximum 1.666133521813222. CG72

computes moving maximum, moving mean, and moving minimum envelopes over the same window $W = 10000$. Its radius statistics match the filtered radius statistics in (12.2)–(12.3), with the same minimum and maximum.

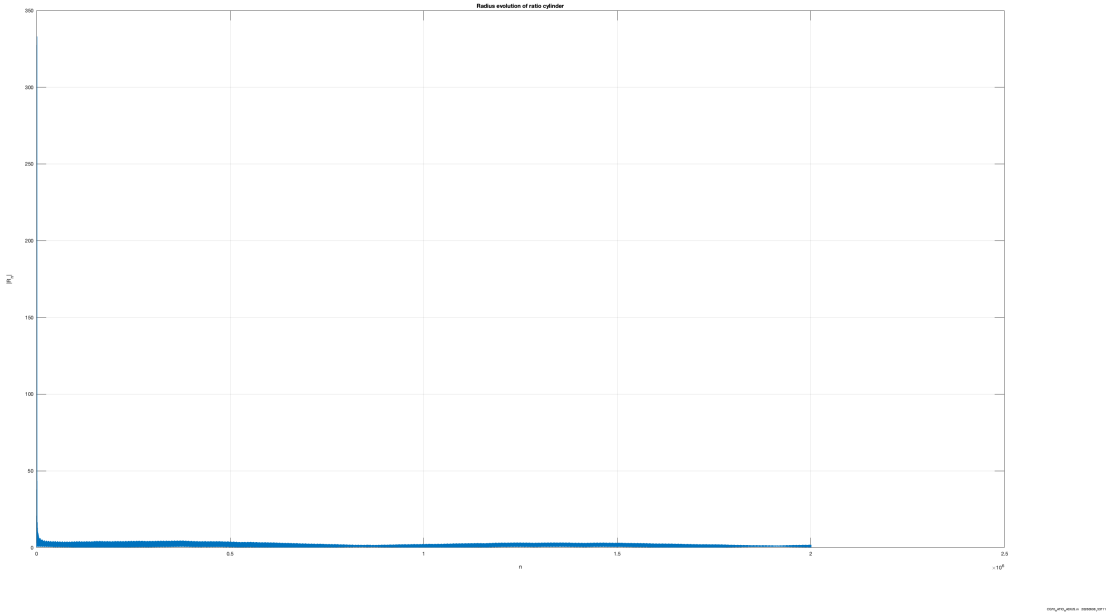


FIGURE 19. Radius evolution $|R_n|$, generated by CG70.

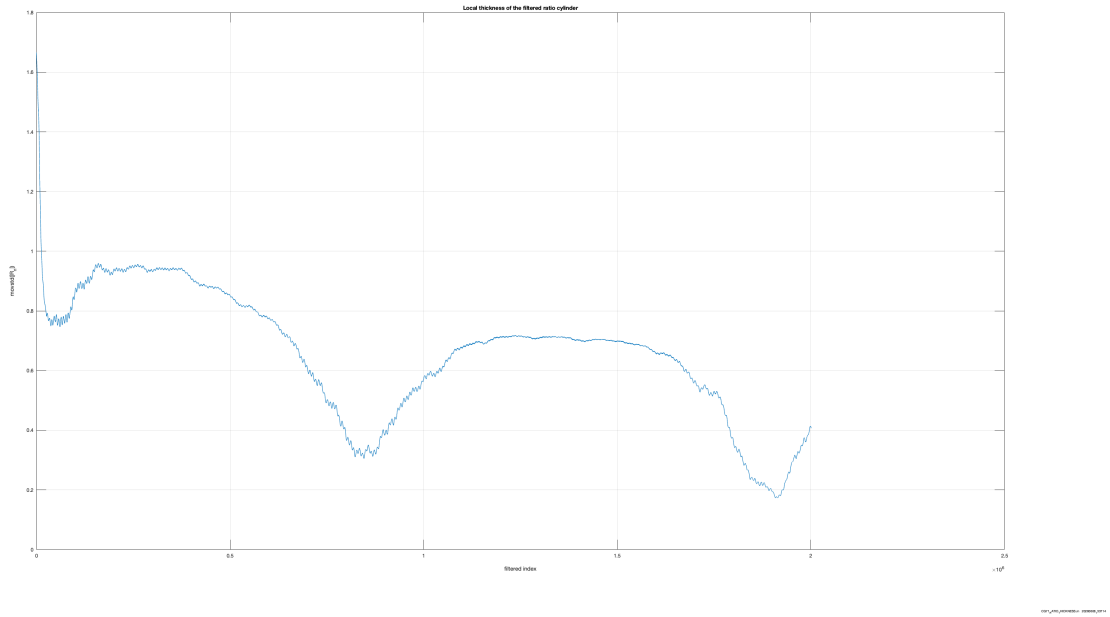


FIGURE 20. Moving local thickness of the filtered ratio cylinder, generated by CG71.

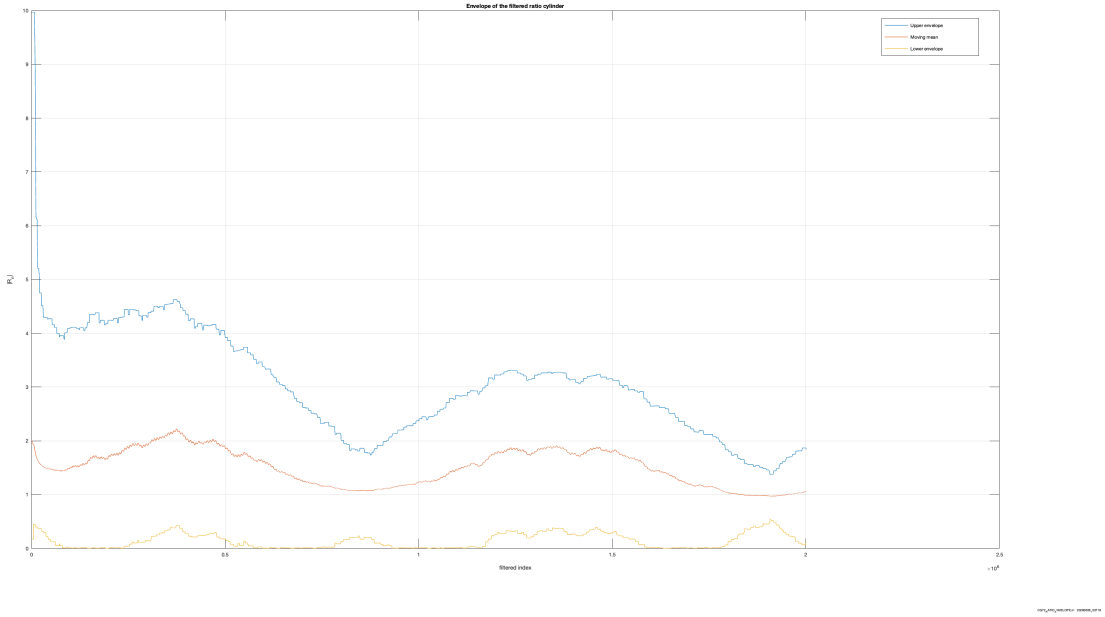


FIGURE 21. Moving envelope of the filtered ratio cylinder, generated by CG72.

14. Notebook Summary

The generated data record a centroid sequence C_n whose first 1000 values have visibly larger coordinate variation than the post-10^4 window. The Lambert centroid sequence $C_n^{(L)}$ has a different cloud location and scale before centering and normalization. After tail-centering and tail-radius normalization, the project compares the two sequences through the normalized difference D_n and ratio R_n. The ratio reports include the full cloud, the full 3D evolution, the last-tenth 3D evolution, and the filtered $|R_n| < 10$ cylinder, together with radius, thickness, and envelope diagnostics. These are numerical observations from the generated CentroidGeometry reports, not proof statements.

References

[1] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford University Press, 1986.

[2] H. M. Edwards, *Riemann's Zeta Function*, Dover Publications, 2001.

[3] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, On the Lambert W function, *Advances in Computational Mathematics* **5** (1996), 329–359, doi:10.1007/BF02124750.